

FILTER DESIGN TEST CASE

Multiplexer Module - n78 (3.3-3.8 GHz 5G NR) for Satellite L-Band Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

Document Number:	FD-TC-0084
Revision:	Rev C.5
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Created Date:	2024-10-10
Last Modified:	2024-11-29
Status:	PASS
Priority:	P1 - Critical

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Multiplexer Module filter targeting n78 (3.3-3.8 GHz 5G NR) for Satellite L-Band Filter. The filter is designed on ScAlN on Si substrate with a target center frequency of 4329.6 MHz and bandwidth of 349.5 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Satellite L-Band Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Multiplexer Module
- Frequency Band: n78 (3.3-3.8 GHz 5G NR)
- Application: Satellite L-Band Filter
- Substrate: ScAlN on Si
- Package: Flip-Chip BGA
- Design Tool: COMSOL Multiphysics 6.2
- Design Center: Osaka Design Center

This specification applies to all variants of the Multiplexer Module design targeting n78 (3.3-3.8 GHz 5G NR). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	4329.6 MHz
Bandwidth (3 dB):	349.5 MHz
Insertion Loss Target:	≤ 2.4 dB
Return Loss Target:	≥ 14.0 dB
Out-of-Band Rejection:	≥ 41.0 dB
Isolation (Duplexer):	≥ 18.0 dB
Q Factor:	1058
Temperature Coefficient:	-8.3 ppm/°C
Die Size:	1.1 x 0.8 mm
Wafer Size:	6-inch
Number of Resonators:	11
Electrode Material:	Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open COMSOL Multiphysics 6.2 and load the Multiplexer Module template project.
2. Define the substrate stack: ScAlN on Si with specified layer thicknesses.
3. Set design targets: center freq 4329.6 MHz, BW 349.5 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.4 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, ScAlN on Si).
10. Perform on-wafer probing using Rohde & Schwarz FSW Signal Analyzer.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 12989 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.4 dB
- Return loss in passband shall be >= 14.0 dB
- Out-of-band rejection at +/- 524 MHz offset >= 41.0 dB
- Group delay variation across passband shall be <= 4 ns
- Temperature drift over -40°C to +85°C shall be <= 4.5 MHz
- Power handling shall be >= +25 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 94%

6. TEST RESULTS

Measurement Date: 2024-11-29
Devices Tested: 53
Insertion Loss (meas): 2.07 dB
Return Loss (meas): 15.5 dB
Rejection (meas): 43.6 dB
Isolation (meas): 23.1 dB
Q Factor (meas): 1058
Group Delay Variation: 3.9 ns
Power Handling: +32 dBm
Die Yield: 84.8%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The Multiplexer Module design demonstrated excellent performance for n78 (3.3-3.8 GHz 5G NR).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Temperature stability was within specification across full range.
- e) Cross-talk between adjacent filter channels was below -55 dB.

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

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Approver:	Dr. James Liu	Date:	2024-12-10

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